



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**UNIVERSITI TEKNOLOGI MALAYSIA
FAKULTI SAINS**

**FINAL EXAMINATION
SEMESTER III**

COURSE CODE : FSPC 0034

COURSE NAME : CHEMISTRY II

PROGRAMME : ASASI UTMSPACE

DATE : 24th APRIL 2022

DURATION : 1 HOUR 45 MINUTES

INSTRUCTION TO STUDENTS:

- 1. THE PAPER CONSISTS OF 3 QUESTIONS. ANSWER ALL QUESTIONS.**
- 2. ALL ANSWERS HAVE TO BE WRITTEN IN YOUR OWN PIECE/S OF PAPER USING BLUE/BLACK-COLOURED PEN.**
- 3. PLEASE WRITE YOUR NAME AND MATRIC CARD NUMBER ON YOUR ANSWER SHEET.**
- 4. PLEASE WRITE THE QUESTION NUMBER CLEARLY ON THE PAPER WHERE YOU ARE ANSWERING.**

(THIS BOOKLET CONSISTS OF 8 PRINTED PAGES INCLUDING THIS PAGE)

QUESTION 4 (10 MARKS)

- (a) In a reaction, 250 mL of 0.3 M bromic acid, HBrO_3 (weak acid), is reacted with 130 mL of 0.2 M sodium hydroxide, NaOH (strong base).

- (i) Determine the pH of the solution given the K_a of bromic acid is 4.5×10^{-7} .
- (ii) Determine whether the salts produced will be acidic, basic or neutral.

(4+1 marks)

- (b) Predict whether the following solutions are basic, acidic or neutral and determine the pH of the solutions.

- (i) 0.2 M KClO_4 (aq)
- (ii) 0.2 M KNO_2 (aq); Given K_a of HNO_2 is 4.5×10^{-4}

(1+4 marks)



B 0.075 0.075 0

A -0.075 0.075 +0.075

A 0 0 0.075

$$[\text{BrO}_3^-] = \frac{0.075}{0.25 + 0.0375} = 0.261 \text{ M}$$



0.261 0 0

-x +x +x

0.261-x x x

$$\text{mol of HBrO}_3 = 0.25 \times 0.3 = 0.075 \text{ mol}$$

$$M_a V_a = M_b V_b$$

$$0.075 = 0.2 V_b, V_b \text{ used} = 0.0375 \text{ L} = 37.5 \text{ mL}$$

$$K_b = \frac{1 \times 10^{-14}}{4.5 \times 10^{-7}} = 2.22 \times 10^{-8}$$

$$2.22 \times 10^{-8} = \frac{x^2}{0.261}$$

$$x = 7.612 \times 10^{-5} \text{ M}$$

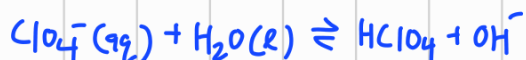
$$\text{pOH} = -\log(7.612 \times 10^{-5})$$

$$= 4.11$$

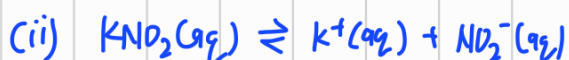
$$\text{pH} = 9.89$$

(ii) basic salts

c. b. a. s.



$\therefore \text{OH}^-$ ion produced, solution is basic



0.2 0 0

-x x x

$$K_b = \frac{1 \times 10^{-14}}{4.5 \times 10^{-4}} = 2.22 \times 10^{-11}$$

$$2.22 \times 10^{-11} = \frac{x^2}{0.2}$$

$$x = 2.107 \times 10^{-6}$$

$$\text{pOH} = -\log(2.107 \times 10^{-6}) = 5.68$$

$$\text{pH} = 9.32$$

QUESTION 5 (10 MARKS)

- a. Differentiate between electrolytic cell and galvanic cell by stating TWO (2) points.

(2 marks)

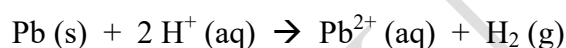
- b. Sketch a simple galvanic cell (in standard condition, inert electrodes). Choose ANY electrodes and electrolyte/solution in your illustration.

(2 marks)

- c. Balance its half reactions to produce a net equation for your chosen cell in (2).

(1 marks)

- d. A voltaic cell consists of a platinum cathode with gaseous hydrogen bubbling over it at a pressure of 1 atm and a lead anode immersed in a 1.00 M solution of $\text{Pb}(\text{NO}_3)_2$. The platinum cathode is submerged in a 0.0013 M of hydrogen ions aqueous solution. Given the net equation for this cell is :

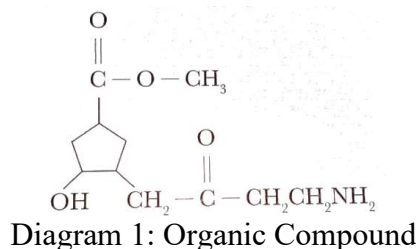


- (i) Write the shorthand notation for this voltaic cell.
- (ii) Calculate the cell potential, E_{cell} .
- (iii) State the spontaneity of the reaction.

(1+3+1 marks)

QUESTION 6 (20 MARKS)

- a. Diagram 1 shows a structure of an organic compound. Please answer the following questions:



For said compound:

- (i) There are more than one functional group. Redraw Diagram 1 in your answer sheet. Identify all the functional groups exist in the compound by drawing a circle around the functional groups.
- (ii) List down all the functional groups and write the names of the class of compound for each functional group.

(2+4 marks)

- b. Give the requirements for a compound to exhibit cis-trans isomerism

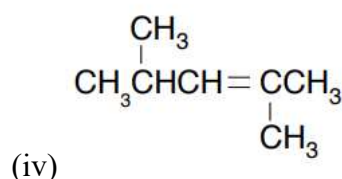
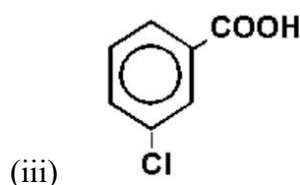
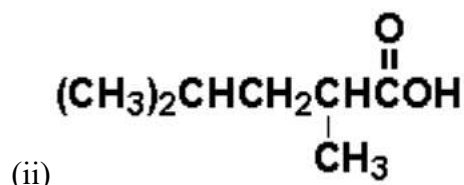
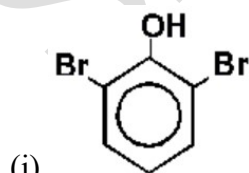
(2 marks)

- c. Determine the type of chemical reaction in each of the following:

- (i) $\text{CH}_3\text{C}\equiv\text{CH} + \text{HCl} \rightarrow \text{CH}_3\text{CCl}=\text{CH}_2$
- (ii) $(\text{CH}_3)_2\text{C}=\text{O} + \text{HCN} \rightarrow (\text{CH}_3)_2\text{C}(\text{OH})\text{CN}$
- (iii) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br} \rightarrow \text{CH}_3\text{CH}=\text{CH}_2 + \text{HBr}$
- (iv) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br} + \text{NaCN} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{CN} + \text{NaBr}$

(4 marks)

- d. Write the name of the following compounds according to the IUPAC Nomenclature.



(4 marks)

- 6) (c) (i) electrophilic addition
(ii) nucleophilic addition
(iii) elimination
(iv) nucleophilic substitution

e. Draw the structures for the following compounds:

- (i) Ethanal
- (ii) Butan-2-ol
- (iii) 1,1-dichloroethene
- (iv) 1,2-dibromocyclohexane

(4 marks)

SPACE UTM

1		18	
IA		VIIIA	
1	2	13	2
H	He	B	He
1.01	4.00	10.81	4.00
3	10	14	9
Li	Ne	C	F
6.94	20.18	12.01	19.00
11	18	15	16
Na	Ar	N	O
22.99	39.95	14.01	16.00
19	36	16	17
K	Kr	P	Cl
39.1	83.80	30.97	35.45
37	54	17	18
Rb	Xe	S	Ar
85.47	131.29	32.07	39.95
55	126.9	34	35
Cs	I	Se	Br
132.9	126.9	78.96	79.90
87	86	52	53
Fr	Rn	Te	At
(223)	(222)	(209)	(210)
		14	15
		IVA	VA
		6	7
		C	N
		12.01	14.01
		13	14
		IIIA	IVA
		5	6
		B	C
		10.81	12.01
		13	14
		Al	Si
		26.98	28.09
		31	32
		Ga	Ge
		69.72	72.61
		49	50
		In	Sn
		114.82	118.71
		81	82
		Tl	Pb
		204.4	207.2
		83	84
		Bi	Po
		209	(209)
		51	52
		Sb	Te
		121.76	127.6
		85	86
		At	Rn
		(210)	(222)
		11	12
		IB	IIB
		29	30
		Cu	Zn
		63.55	65.39
		47	48
		Ag	Cd
		107.87	112.41
		79	80
		Au	Hg
		197.0	200.6
		111	
		Rg	
		(272)	
		10	11
		10	11
		Ni	Cu
		58.69	63.55
		46	47
		Pd	Ag
		106.42	107.87
		78	79
		Pt	Au
		195.1	197.0
		110	
		Ds	
		(271)	(272)
		9	10
		9	10
		Co	Ni
		58.93	58.69
		45	46
		Rh	Pd
		102.91	106.42
		77	78
		Ir	Pt
		192.2	195.1
		109	110
		Mt	Ds
		(268)	(271)
		8	9
		8	9
		Fe	Co
		55.85	58.93
		44	45
		Ru	Rh
		101.07	102.91
		76	77
		Os	Ir
		190.2	192.2
		108	109
		Hs	Mt
		(265)	(268)
		7	8
		7	8
		Mn	Fe
		54.94	55.85
		43	44
		Tc	Ru
		(98)	101.07
		75	76
		Re	Os
		186.2	190.2
		107	108
		Bh	Hs
		(264)	(265)
		6	7
		6	7
		Cr	Mn
		52.00	54.94
		42	43
		Mo	Tc
		95.94	(98)
		74	75
		W	Re
		183.9	186.2
		106	107
		Sg	Bh
		(263)	(264)
		5	6
		5	6
		V	Cr
		50.94	52.00
		41	42
		Nb	Mo
		92.91	95.94
		73	74
		Ta	W
		180.9	183.9
		105	106
		Db	Sg
		(262)	(263)
		4	5
		4	5
		Ti	V
		47.88	50.94
		40	41
		Zr	Nb
		91.22	92.91
		72	73
		Hf	Ta
		178.5	180.9
		104	105
		Rf	Db
		(261)	(262)
		3	4
		3	4
		Sc	Ti
		44.96	47.88
		39	40
		Y	Zr
		88.91	91.22
		57	58
		La*	Ba
		138.9	137.3
		89	88
		Ac^	Ra
		(227)	(226)
		12	13
		IIA	IIIA
		4	5
		Be	B
		9.01	10.81
		20	21
		Ca	Sc
		40.08	44.96
		38	39
		Sr	Y
		87.62	88.91
		56	57
		Ba	La*
		137.3	138.9
		88	89
		Ra	Ac^
		(226)	(227)
		60	61
		Nd	Pm
		144.2	(145)
		92	93
		U	Np
		238.0	(237)
		91	92
		Pa	U
		(231)	238.0
		58	59
		Ce	Pr
		140.1	140.9
		90	91
		Th	Pa
		232.0	(231)
		62	63
		Sm	Eu
		150.4	152.0
		94	95
		Pu	Am
		(244)	(243)
		64	65
		Gd	Tb
		157.3	158.9
		96	97
		Cm	Bk
		(247)	(247)
		66	67
		Dy	Ho
		162.5	164.9
		98	99
		Cf	Es
		(251)	(252)
		70	71
		Yb	Lu
		173.0	175.0
		102	103
		No	Lr
		(259)	(260)

*

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LIST OF SELECTED CONSTANT VALUES

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$
		=	$2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
		=	$0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	μ_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	β_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

UNIT AND CONVERSION FACTOR

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

STANDARD REDUCTION POTENTIAL (E°_{SRP})

Half-Reaction	E°_{red}
$\text{K}^{+} + \text{e}^{-} \rightleftharpoons \text{K}$	-2.924
$\text{Ba}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Ba}$	-2.900
$\text{Ca}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Ca}$	-2.760
$\text{Na}^{+} + \text{e}^{-} \rightleftharpoons \text{Na}$	-2.712
$\text{Mg}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Mg}$	-2.375
$\text{H}_2 + 2\text{e}^{-} \rightleftharpoons 2\text{H}^{-}$	-2.230
$\text{Al}^{3+} + 3\text{e}^{-} \rightleftharpoons \text{Al}$	-1.706
$\text{Mn}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Mn}$	-1.040
$\text{Zn}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Zn}$	-0.763
$\text{Cr}^{3+} + 3\text{e}^{-} \rightleftharpoons \text{Cr}$	-0.740
$\text{S} + 2\text{e}^{-} \rightleftharpoons \text{S}^{2-}$	-0.508
$2\text{CO}_2 + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{H}_2\text{C}_2\text{O}_4$	-0.490
$\text{Cr}^{3+} + \text{e}^{-} \rightleftharpoons \text{Cr}^{2+}$	-0.410
$\text{Fe}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Fe}$	-0.409
$\text{Co}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Co}$	-0.280
$\text{Ni}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Ni}$	-0.230
$\text{Sn}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Sn}$	-0.136
$\text{Pb}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Pb}$	-0.126
$\text{Fe}^{3+} + 3\text{e}^{-} \rightleftharpoons \text{Fe}$	-0.036
$2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{H}_2$	0.000
$\text{S}_4\text{O}_6^{2-} + 2\text{e}^{-} \rightleftharpoons 2\text{S}_2\text{O}_3^{2-}$	0.089
$\text{Sn}^{4+} + 2\text{e}^{-} \rightleftharpoons \text{Sn}^{2+}$	0.150
$\text{Cu}^{2+} + \text{e}^{-} \rightleftharpoons \text{Cu}^{+}$	0.158
$\text{Cu}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Cu}$	0.340
$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^{-} \rightleftharpoons 4\text{OH}^{-}$	0.401
$\text{Cu}^{+} + \text{e}^{-} \rightleftharpoons \text{Cu}$	0.522
$\text{I}_3^{-} + 2\text{e}^{-} \rightleftharpoons 3\text{I}^{-}$	0.534
$\text{MnO}_4^{-} + 2\text{H}_2\text{O} + 3\text{e}^{-} \rightleftharpoons \text{MnO}_2 + 4\text{OH}^{-}$	0.588
$\text{O}_2 + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{H}_2\text{O}_2$	0.682
$\text{Fe}^{3+} + \text{e}^{-} \rightleftharpoons \text{Fe}^{2+}$	0.770
$\text{Hg}_2^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Hg}$	0.796
$\text{Ag}^{+} + \text{e}^{-} \rightleftharpoons \text{Ag}$	0.800
$\text{Hg}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Hg}$	0.851
$\text{H}_2\text{O}_2 + 2\text{e}^{-} \rightleftharpoons 2\text{OH}^{-}$	0.880
$\text{HNO}_3 + 3\text{H}^{+} + 3\text{e}^{-} \rightleftharpoons \text{NO} + 2\text{H}_2\text{O}$	0.960
$\text{Br}_2(\text{aq}) + 2\text{e}^{-} \rightleftharpoons 2\text{Br}^{-}$	1.087
$2\text{IO}_3^{-} + 12\text{H}^{+} + 10\text{e}^{-} \rightleftharpoons \text{I}_2 + 6\text{H}_2\text{O}$	1.190
$\text{CrO}_4^{2-} + 8\text{H}^{+} + 3\text{e}^{-} \rightleftharpoons \text{Cr}^{3+} + 4\text{H}_2\text{O}$	1.195
$\text{Pt}^{2+} + 2\text{e}^{-} \rightleftharpoons \text{Pt}$	1.200
$\text{MnO}_2 + 4\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{Mn}^{2+} + 2\text{H}_2\text{O}$	1.208
$\text{O}_2 + 4\text{H}^{+} + 4\text{e}^{-} \rightleftharpoons 2\text{H}_2\text{O}$	1.229
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^{+} + 6\text{e}^{-} \rightleftharpoons 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.330
$\text{Cl}_2(\text{g}) + 2\text{e}^{-} \rightleftharpoons 2\text{Cl}^{-}$	1.358
$\text{PbO}_2 + 4\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{Pb}^{2+} + 2\text{H}_2\text{O}$	1.467
$\text{MnO}_4^{-} + 8\text{H}^{+} + 5\text{e}^{-} \rightleftharpoons \text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.491
$\text{Au}^{+} + \text{e}^{-} \rightleftharpoons \text{Au}$	1.680
$\text{H}_2\text{O}_2 + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons 2\text{H}_2\text{O}$	1.776
$\text{Co}^{3+} + \text{e}^{-} \rightleftharpoons \text{Co}^{2+}$	1.842
$\text{S}_2\text{O}_8^{2-} + 2\text{e}^{-} \rightleftharpoons 2\text{SO}_4^{2-}$	2.050
$\text{O}_3(\text{g}) + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons \text{O}_2(\text{g}) + \text{H}_2\text{O}$	2.070
$\text{F}_2(\text{g}) + 2\text{H}^{+} + 2\text{e}^{-} \rightleftharpoons 2\text{HF}(\text{aq})$	3.030